Adaptive Risk-Aware Reinforcement Learning for Vision-Based Indoor Drone Navigation

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Research-status note: This is an original synthetic example. Numerical results, datasets, experiments, and performance values are illustrative and are not empirical findings from a completed study.

Abstract— This paper presents a risk-aware reinforcement learning framework for autonomous indoor quadrotor navigation using monocular depth observations and a compact state vector. The proposed agent combines a convolutional encoder with a multilayer state encoder and employs a risk-sensitive reward formulation separating progress, collision avoidance, heading alignment, and control smoothness. A curriculum mechanism gradually increases obstacle density during training. An illustrative simulation study compares a PPO agent using the proposed reward against simpler baselines. The results are synthetic examples showing how training stability and navigation safety can be evaluated.

Keywords: reinforcement learning; autonomous drones; PPO; depth perception; reward shaping; indoor navigation; curriculum learning

1. Introduction

Autonomous indoor flight requires an agent to reason about obstacles, goal geometry, and vehicle motion under partial observability. Reinforcement learning (RL) is attractive because navigation can be formulated as sequential decision making, but training is sensitive to reward scale and task difficulty. A large collision penalty can dominate small progress rewards and encourage an overly conservative policy.

This paper presents a synthetic study of a risk-aware PPO navigation framework using depth images and a compact state vector. The design combines multimodal feature extraction, bounded reward components, and curriculum learning. The purpose is to demonstrate a complete research-paper structure and an evaluation methodology.

2. Related Work

Deep Q-Networks established neural value approximation for high-dimensional control, while Double DQN and dueling architectures addressed value overestimation and action-value decomposition. PPO introduced a clipped policy objective that is widely used for stable on-policy optimization. In robotics, simulation is frequently used to collect large numbers of interactions before physical deployment.

3. Methodology

3.1 Observation and Policy

The observation is $o_t = \{I_t, s_t\}$, where I_t is a depth image and s_t contains normalized position, velocity, goal distance, and heading information. A convolutional encoder processes the image, while a multilayer perceptron processes the state vector. Their embeddings are concatenated before policy and value heads produce the control distribution and state-value estimate.

3.2 Risk-Aware Reward

The illustrative reward is $R_t = 4\Delta d_t + 1.5A_t - 0.2 - 8C_t - 3O_t$. Here Δd_t is normalized progress, A_t is heading alignment, C_t indicates collision, and O_t indicates leaving the allowed operating region. Continuous terms are clipped before combination so that a single component cannot dominate the optimization.

3.3 Curriculum

Training is divided into three stages. Stage 1 uses sparse obstacles and short goal distances, Stage 2 increases obstacle density and distance, and Stage 3 samples from the full task distribution. Advancement is triggered by a rolling success-rate threshold.

4. Experimental Setup

The synthetic environment is a 5 m × 20 m indoor corridor simulated with a PyBullet-style physics engine. The vehicle is controlled at 30 Hz while the learned policy is queried at 10 Hz. Five random seeds are used per configuration. Each training run contains 500,000 environment steps.

Configuration	Observation	Reward	Curriculum
PPO-Risk	Depth + state	Risk-aware	Yes
PPO-Base	Depth + state	Unshaped	No
DQN-Discrete	Depth + state	Risk-aware	No

Table 1. Illustrative experimental configurations.

4.1 Metrics

Evaluation uses success rate, collision rate, mean episode return, path efficiency, and return variance. Success means entering the goal region without collision before the episode horizon.

5. Results

Method	Success (%)	Collision (%)	Path efficiency	Return
PPO-Risk	84.2	9.8	0.78	61.4
PPO-Base	69.5	18.7	0.70	43.1
DQN-Discrete	62.1	23.4	0.65	37.8

Table 2. Illustrative evaluation results averaged over five synthetic runs.

The synthetic results show the reporting pattern expected in a completed experiment: the risk-aware configuration has a higher success rate and lower collision rate, while the baseline configurations trade some safety for different learning behavior. These values are examples only and do not establish empirical superiority.

5.1 Training Stability

The illustrative learning curves are consistent with a curriculum reducing variance during transitions to denser obstacle layouts. A complete empirical paper should report confidence intervals and statistical tests rather than relying only on mean curves.

6. Discussion

Reward scaling is a central design issue. If a collision penalty is orders of magnitude larger than typical progress rewards, the optimizer may learn that inactivity is safer than exploration. Conversely, a weak safety penalty can produce aggressive trajectories. Logging each reward component separately provides a practical way to diagnose this imbalance.

Combining visual and low-dimensional observations also separates two information sources: image features encode local geometry, while explicit state variables provide accurate goal-relative information. Real deployment would additionally require sensor-noise modeling, domain randomization, and hardware validation.

7. Limitations and Future Work

This manuscript uses synthetic experimental values and does not claim a real-world benchmark. Future work should replace them with reproducible experiments, multiple independent seeds, confidence intervals, sensor corruption tests, and sim-to-real evaluation. Recurrent policies and constrained RL are also promising extensions.

8. Conclusion

A risk-aware PPO framework for indoor drone navigation was described using depth observations, state variables, bounded reward components, and curriculum learning. The synthetic study illustrates how success, collision, efficiency, and return can be reported together while keeping the distinction between methodological proposal and empirical evidence explicit.

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